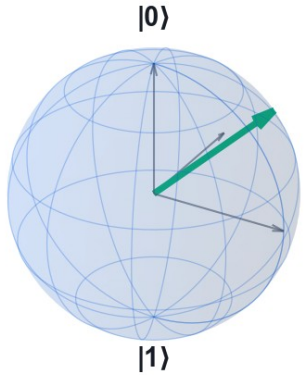



COURSE SLIDES · NON-WSQ

# Quantum Computing Fundamentals for Beginners

Course Code: C882

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 2 October 2026

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COURSE ADMINISTRATION

# Welcome & Housekeeping


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# About the Trainer




## Your Trainer

General Trainer template —
to be completed by the trainer

**NAME**

**TITLE / DESIGNATION**

**QUALIFICATIONS**

**AREAS OF EXPERTISE**

**TRAINING & INDUSTRY EXPERIENCE**

**CONTACT**


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YOUR TRAINER

# About the Trainer




## Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

**ROLE**
Principal Trainer, Tertiary Infotech Academy Pte Ltd

**QUALIFICATIONS**
PhD; ACTA / DACE certified adult educator.

**EXPERTISE**
Quantum computing, quantum algorithms, machine learning and data analytics.

**DELIVERS**
Courses on quantum computing, quantum cryptography, Python and AI.

**FOUNDER**
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.


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ICE-BREAKER

# Let's Know Each Other

1

## Your name and organisation

Tell us your role and what your team does.

2

## Your background

Any physics, mathematics or programming experience — none is required.

3

## Why quantum

What drew you to quantum computing?

4

## Your goal

What do you want to be able to build by the end of Day 2?


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HOUSEKEEPING

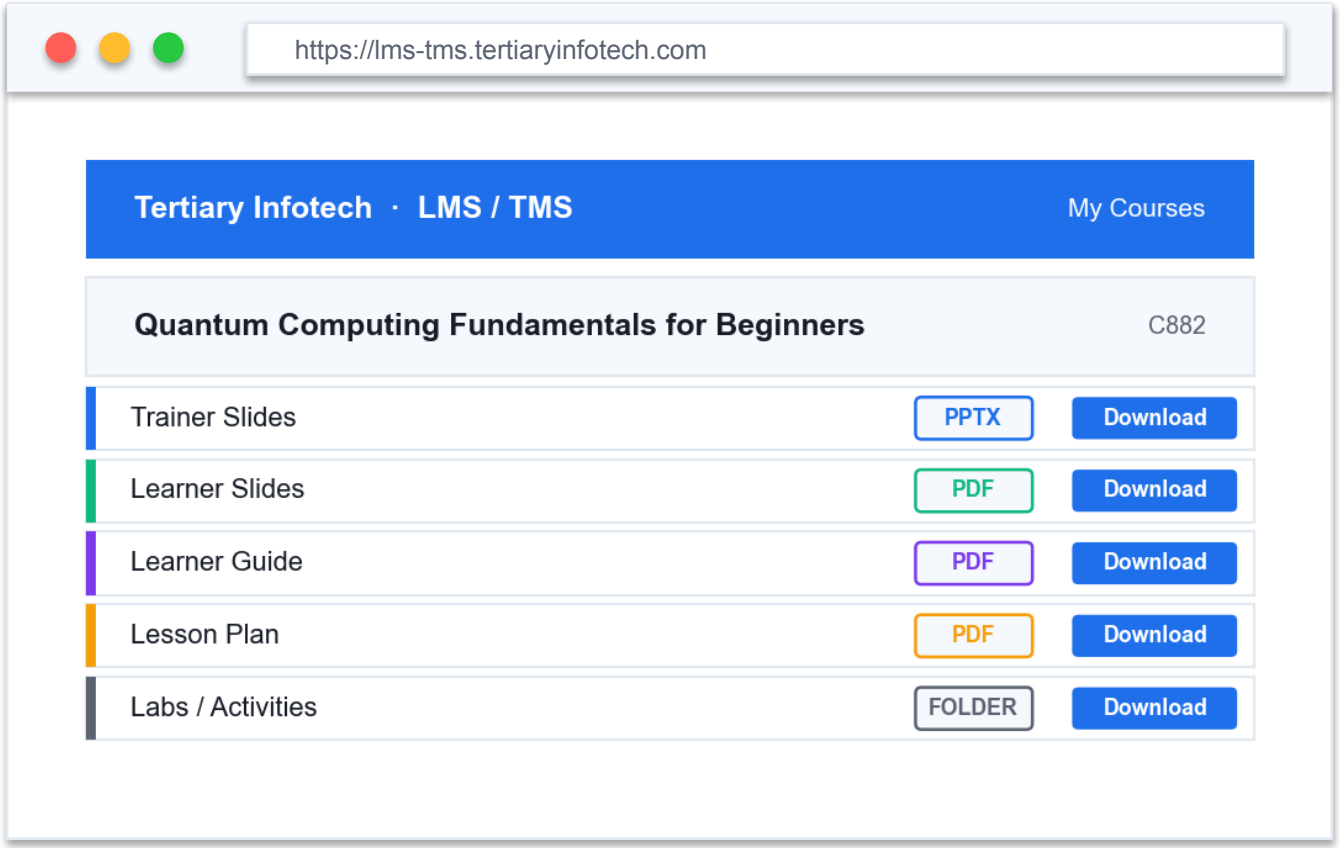
# Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Respect each other's views; agree to disagree.
4. Exit quietly for a phone call or toilet break.
5. Be punctual; return from breaks on time.
6. Full attendance — attend both days to get the most from the course.


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LMS / TMS · COURSE PORTAL

# Download Your Course Material




1. Open https://lms-tms.tertiaryinfotech.com in your browser.
2. Sign in with the account e-mailed to you before the course.
3. Open this course: Quantum Computing Fundamentals for Beginners.
4. Download the Learner Slides, Learner Guide and Lab files.
5. Keep them open — you will refer to them in every lab.

Download everything at the start of Day 1 — you will use them throughout the course.


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SCHEDULE · 15 INSTRUCTIONAL HOURS

# Lesson Plan — 2 Days, 7.5 hours per day

## Day 1 · 9:30am – 5:30pm

- Day 1 — Quantum Principles, States and Gates
  - Welcome · Introductions
  - Topic 1: Introduction to Quantum Computing (Labs 1–3)
  - Lunch Break
  - Topic 2: Quantum States and Gates (Labs 4–7)
  - End of Day 1

## Day 2 · 9:30am – 5:30pm

- Day 2 — Quantum Algorithms and Applications
  - Day 1 Recap
  - Topic 3: Quantum Algorithms (Labs 8–13)
  - Lunch Break
  - Topic 4: Quantum Computing Applications (Labs 14–17)
  - Course Summary · Q&A · Course Feedback


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FOUR TOPICS · 17 HANDS-ON LABS

# Course Outline

1

## Topic 1 · Introduction to Quantum Computing

Key principles of quantum computing; quantum computing platforms.

2

## Topic 2 · Quantum States and Gates

Quantum state vector representations; quantum circuits and quantum gates.

3

## Topic 3 · Quantum Algorithms

Coding quantum algorithms; testing quantum algorithms.

4

## Topic 4 · Quantum Computing Applications

Introduction to quantum error correction; introduction to quantum machine learning.


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01

TOPIC 01

# Introduction to Quantum Computing

Key Principles of Quantum Computing · Quantum Computing Platforms


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WHAT IS A QUANTUM COMPUTER

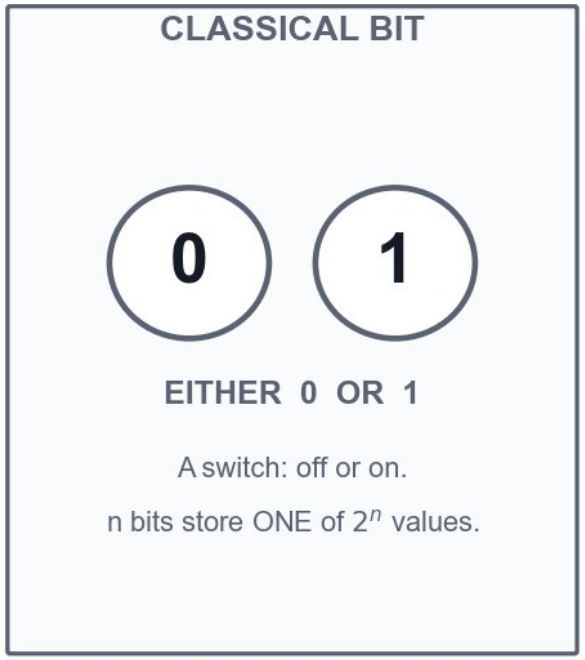
# A quantum computer computes with superposition and entanglement.

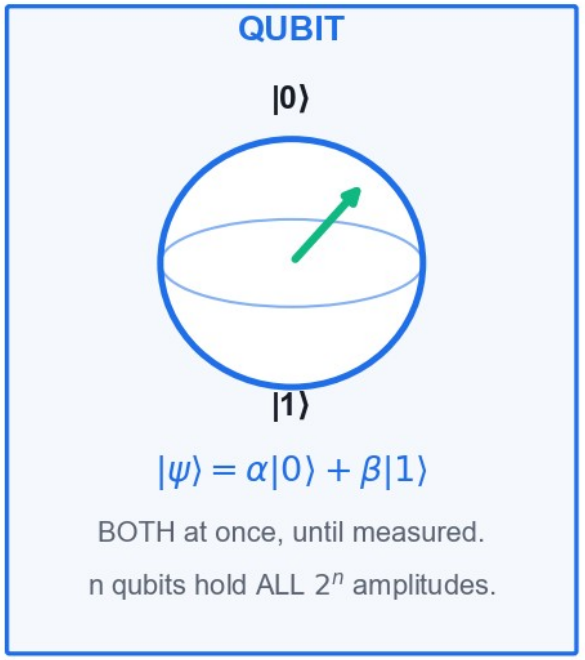
It performs calculations on qubits using two laws of the quantum-mechanical world that have no classical equivalent.


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THE FUNDAMENTAL UNIT

# Classical Bit vs Qubit




### Classical bit

Exactly 0 or 1 — a switch that is either off or on. n bits store ONE of 2^n values.

### Qubit

|0>, |1>, or a superposition a|0> + b|1> of both at once. n qubits hold ALL 2^n amplitudes.

### Measurement

Collapses the qubit to 0 or 1 with probability |a|^2 and |b|^2 — you never see a and b directly.

### Why it matters

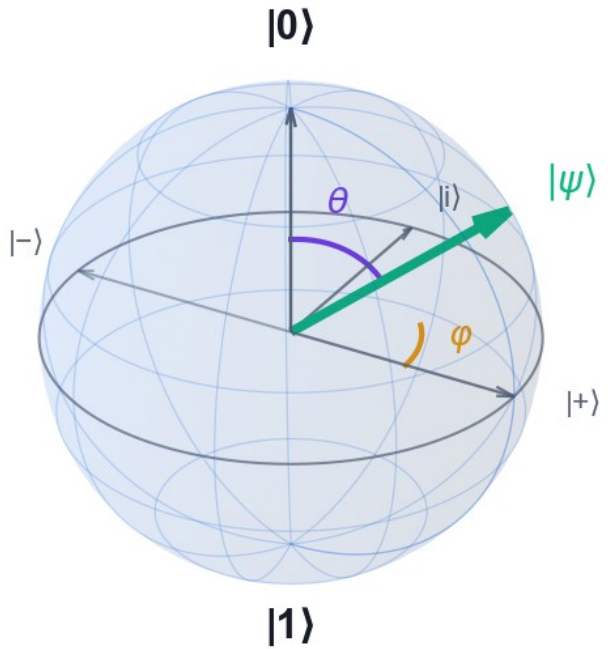
The state space doubles with every qubit added — 50 qubits already exceed a supercomputer.

A qubit is a superposition of |0> and |1>; a bit is one or the other.


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GEOMETRY OF A SINGLE QUBIT

# Visualising a Qubit — The Bloch Sphere




**North pole**

|0> — measures 0 with certainty.

**South pole**

|1> — measures 1 with certainty.

**Equator**

Equal superpositions such as |+> and |-> — a 50/50 measurement.

**Angles**

theta sets the probability; phi sets the relative phase.

|psi> = cos(theta/2)|0> + e^(i.phi) sin(theta/2)|1>


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THE MATHEMATICS

# Quantum State Vector Representation

## Single qubit

```
|psi> = a|0> + b|1>

        |a|^2 + |b|^2 = 1
```

The amplitudes a and b are complex numbers. Their squared magnitudes are the measurement probabilities, and they must sum to 1 (normalisation).

## Dirac notation

```
|0> = [1, 0]^T
        |1> = [0, 1]^T
```

The computational basis states as column vectors. Every gate is a unitary matrix multiplying this vector.

## Two qubits

```
|psi> = a|00> + b|01>
          + c|10> + d|11>
```

n qubits need 2^n amplitudes. This exponential growth is the origin of quantum advantage — and of the difficulty of simulating quantum systems classically.


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HOW WE GOT HERE

# Brief History of Quantum Computing

1

### 1980–82 · The idea

Benioff describes a quantum Turing machine; Feynman argues that simulating quantum physics needs a quantum computer.

2

### 1985 · Deutsch

David Deutsch formalises the universal quantum computer and the first quantum algorithm.

3

### 1994–96 · The algorithms

Shor factors integers exponentially faster; Grover searches quadratically faster.

4

### 2019 · Quantum supremacy

Google's Sycamore completes a sampling task claimed to be intractable classically.

5

### 2020s · NISQ era

IBM, Google, IonQ and others field 50–1,000+ noisy qubits on the cloud.

6

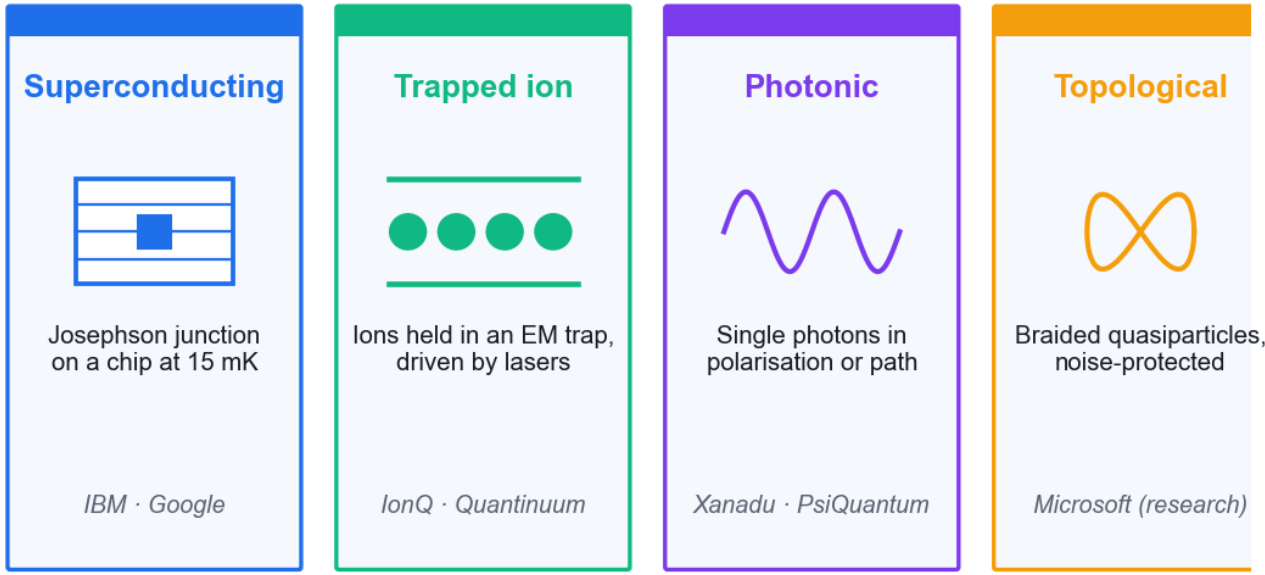
### Today · Toward fault tolerance

Error-corrected logical qubits are demonstrated; useful fault tolerance is the next milestone.


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ANY TWO-LEVEL QUANTUM SYSTEM

# Physical Implementation of a Qubit




Any two-level quantum-mechanical system can be used as a qubit.

**Superconducting**
Josephson-junction circuits at millikelvin temperatures — IBM and Google.

**Trapped ion**
Individual ions held in electromagnetic traps and driven by lasers — IonQ, Quantinuum.

**Photonic**
Single photons carrying the state in polarisation or path — Xanadu, PsiQuantum.

**Topological**
Braided quasiparticles, intrinsically protected from noise — Microsoft, still experimental.

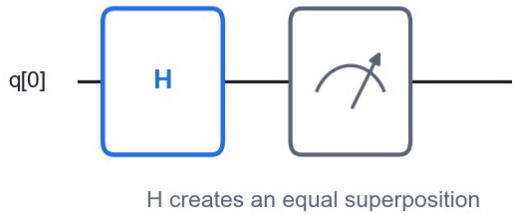
Any two-level quantum-mechanical system can serve as a qubit.


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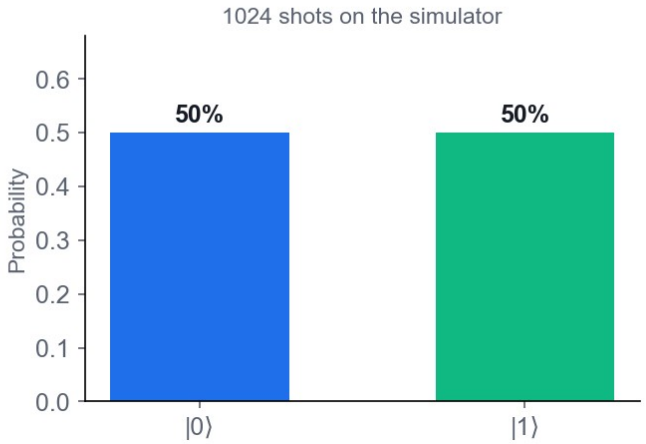
PRINCIPLE 1

# Quantum Superposition

**Superposition: H|0⟩ = (|0⟩ + |1⟩)/√2**




H creates an equal superposition




## Both at once

Before measurement the qubit genuinely holds both |0> and |1> with amplitudes.

## Interference

Amplitudes are complex — they add and CANCEL. Algorithms engineer cancellation of wrong answers.

## Collapse

Measurement destroys the superposition; you get one classical bit, at random.

## Not just probability

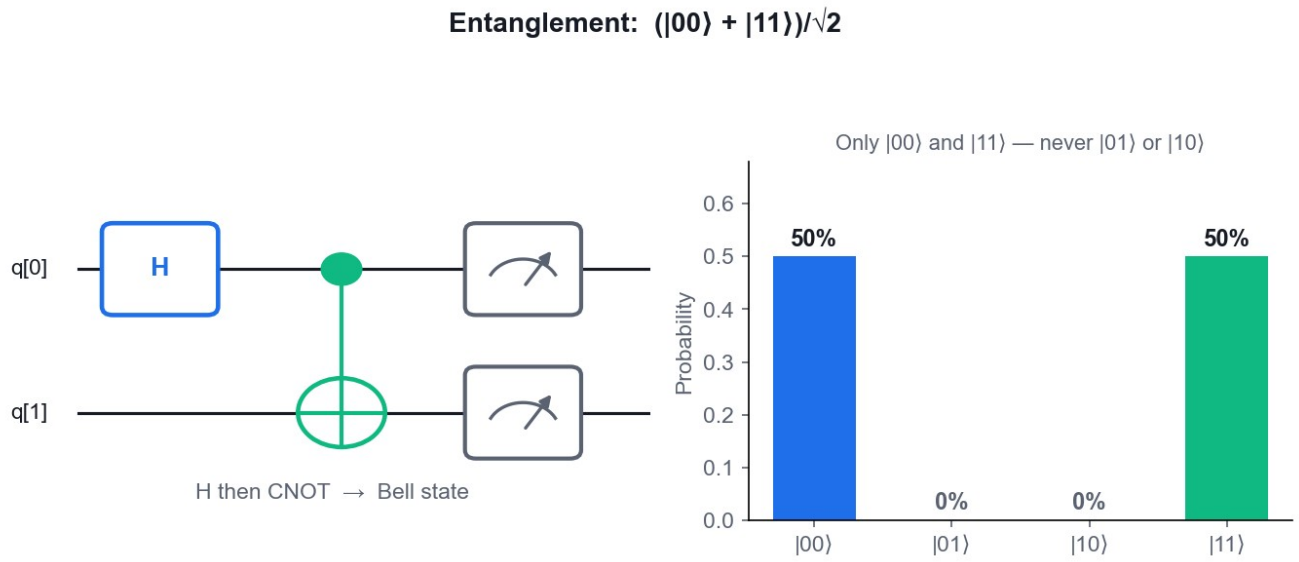
A classical coin is unknown, not undecided. Interference proves the difference.

H|0> = (|0> + |1>)/sqrt(2) — the Hadamard gate creates superposition.


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PRINCIPLE 2

# Quantum Entanglement




The Bell state (|00> + |11>)/sqrt(2) — never |01> or |10>.

## Linked states

Two or more particles whose quantum states cannot be described independently.

## Instant correlation

Measuring one instantly determines the other, at any distance.

## EPR paradox

Einstein, Podolsky and Rosen called it 'spooky action at a distance' and thought QM incomplete.

## Bell's answer

Bell inequalities — and the CHSH experiment — showed nature really is non-local.


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THE SEPARABILITY CONDITION

# Separability — Is a State Entangled?

1

## Product state

If |psi> can be written as |a> tensor |b>, the two qubits are INDEPENDENT — not entangled.

2

## The test

For a|00> + b|01> + c|10> + d|11>, the state is separable if and only if ad = bc.

3

## Worked example

The Bell state has a = d = 1/sqrt(2) and b = c = 0, so ad = 1/2 but bc = 0. They differ — it IS entangled.

4

## Counter-example

(|00> + |01>)/sqrt(2) gives ad = 0 and bc = 0. Equal — so it is NOT entangled; it is just |0> tensor |+>.


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THE RULES OF THE GAME

# The Five Postulates of Quantum Mechanics

1

### 1 · State space

A closed quantum system is described by a unit vector in a complex Hilbert space.

2

### 2 · Time evolution

A closed system evolves by a UNITARY operator: |psi'> = U|psi>. Every quantum gate is unitary, hence reversible.

3

### 3 · Observables

Every measurable quantity is a Hermitian operator; its eigenvalues are the possible results.

4

### 4 · Born rule

The probability of outcome m is |<m|psi>|^2 — the squared amplitude.

5

### 5 · Collapse

After measuring m, the state becomes |m>. The superposition is destroyed.

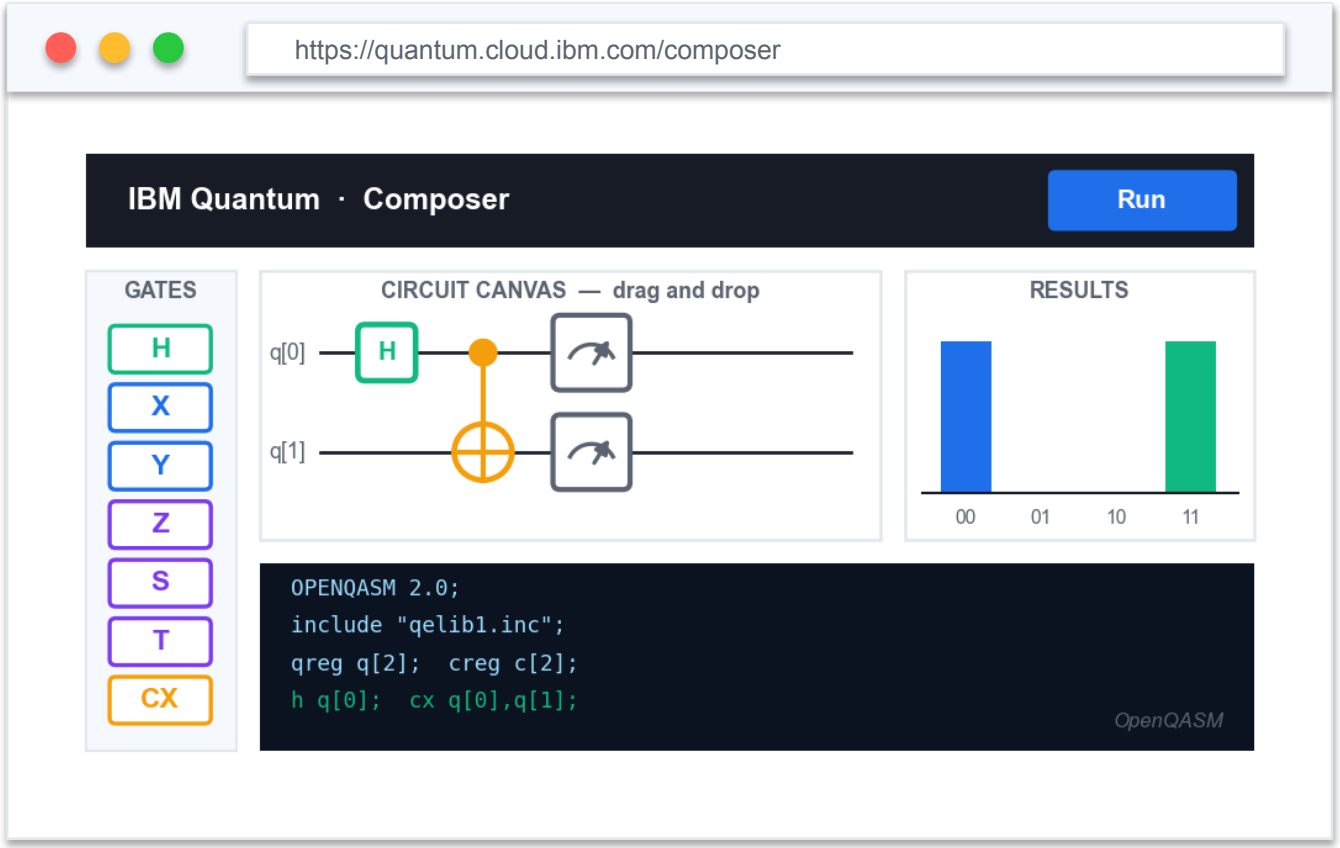
6

### Grouping

Postulates 1–2 describe state and deterministic evolution; 3–5 describe measurement.


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# Quantum Computing Platforms




1. IBM Quantum Composer — drag-and-drop circuits, simulator and real hardware. Our main tool.
2. IBM Quantum Platform — job queue, backends and calibration data.
3. Quirk (algassert.com/quirk) — instant in-browser statevector visualisation.
4. Quantum Circuit Designer — thequantumlaend.de, good for larger Grover circuits.
5. Qiskit and PennyLane — Python SDKs for coding circuits programmatically.

Every lab in this course runs on IBM Quantum Composer — no installation, no cost, works in any browser.


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WHY INDUSTRY CARES

# Use Cases of Quantum Computing

1

## Chemistry & pharma

Simulate molecules exactly — drug discovery, catalysts, nitrogen fixation.

2

## Finance

Portfolio optimisation, derivative pricing and risk analysis by quantum Monte Carlo.

3

## Logistics

Route and fleet optimisation, scheduling and supply-chain planning (QAOA).

4

## Materials science

Design superconductors, batteries and novel materials from first principles.

5

## Cryptography

Shor's algorithm threatens RSA/ECC; QKD offers physics-based key exchange.

6

## Machine learning

Quantum kernels, variational classifiers and quantum-enhanced optimisation.


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WHAT YOU'LL DO ON IBM QUANTUM COMPOSER

# Hands-On Labs — Introduction to Quantum Computing

## Lab 1

- Explore the Qubit on the Bloch Sphere

## Lab 2

- Observe Superposition and Interference

## Lab 3

- Create Your First Entangled Pair (Bell State)


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TOPIC 01 · HANDS-ON

# Explore the Qubit on the Bloch Sphere

**LAB 1**

Bloch Sphere Simulator

Use an interactive Bloch-sphere simulator to see what a qubit actually is: a point on a sphere. Apply X, H and Z and watch the state vector move, so that every gate you meet later has a picture attached to it.

1. Open the Bloch sphere simulator and identify the three…
2. Confirm the qubit starts at the north pole — the state…
3. Apply the X gate
4. Reset, then apply H
5. With the state at |+>, apply Z

**YOU'LL PRODUCE** A worked set of Bloch-sphere states (|0>, |1>, |+>, |->) and the gates that produce them.

**Full step-by-step guide: Learner Guide, Lab 1**


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TOPIC 01 · HANDS-ON

# Observe Superposition and Interference

**LAB 2**

IBM Quantum Composer

Build the smallest possible quantum circuit — a single Hadamard gate — on IBM Quantum Composer, run it on the simulator, and read the probability histogram. Then add a second Hadamard to see interference cancel the superposition completely.

1. Sign in to IBM Quantum and open the Composer from the…
2. Create a new circuit and set the circuit width to 1 qubit…
3. Drag an H gate onto qubit q[0], then drag a Measure…
4. Open the OpenQASM editor and confirm the code reads: h…
5. Run the circuit on the simulator with 1024 shots and read…

**YOU'LL PRODUCE** Two single-qubit circuits: H (a 50/50 random bit) and H-H (a deterministic 0), with their histograms.

**Full step-by-step guide: Learner Guide, Lab 2**


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# Create Your First Entangled Pair (Bell State)

**LAB 3**

IBM Quantum Composer

Build the Bell state (|00> + |11>)/sqrt(2) on IBM Quantum Composer with just H and CNOT, and confirm from the histogram that the two qubits are perfectly correlated — the signature of entanglement.

1. Open a new circuit in IBM Quantum Composer with 2 qubits…
2. Place an H gate on q[0]
3. Place a CNOT with the control on q[0] and the target on…
4. Add Measure operations on both q[0] and q[1]
5. Inspect the statevector panel: the amplitudes are…

**YOU'LL PRODUCE** A 2-qubit Bell-state circuit whose histogram shows only |00> and |11>, each near 50%.

**Full step-by-step guide: Learner Guide, Lab 3**


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# Lunch Break

30 minutes

TOPIC 01 RECAP

# Recap — Introduction to Quantum Computing

- A qubit is a superposition of |0> and |1>, drawn as a point on the Bloch sphere.
- Superposition plus interference is what makes quantum algorithms faster — not randomness alone.
- Entanglement links qubits so that measuring one determines the other; test separability with ad = bc.
- Any two-level quantum system works as a qubit — superconducting, trapped ion, photonic or topological.
- IBM Quantum Composer gives you a free simulator and real hardware in the browser.


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02

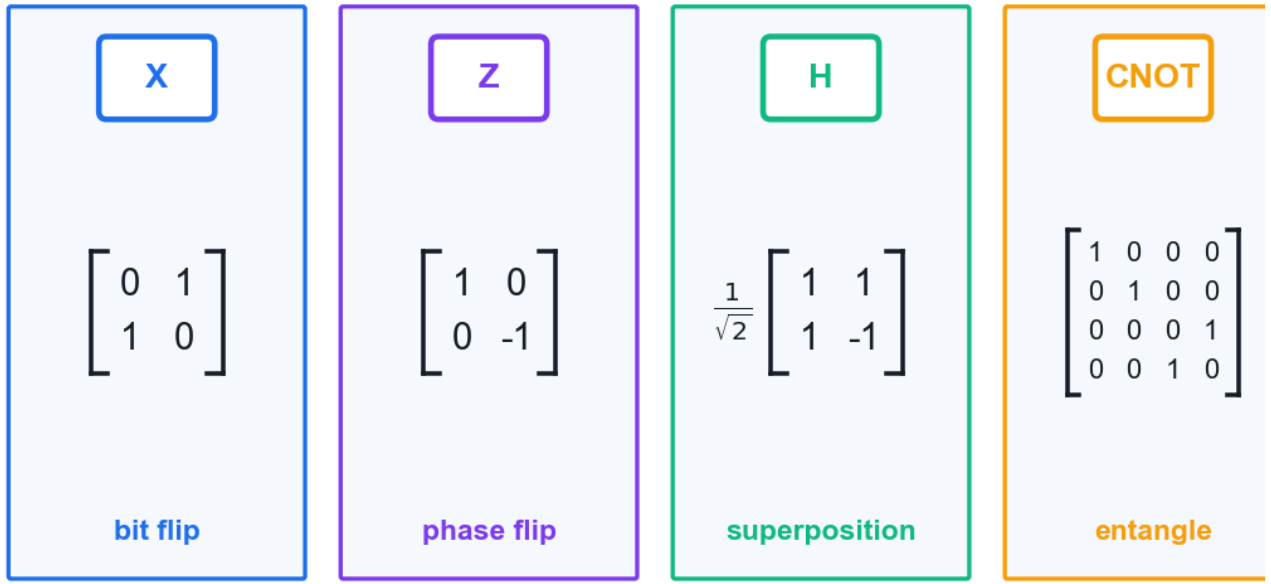
TOPIC 02

# Quantum States and Gates

Quantum State Vector Representations · Quantum Circuits and Quantum Gates


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# Quantum Gates Are Unitary Matrices




Every gate is unitary: $UU^{\dagger} = I$ — so every quantum operation is reversible.

Gates rotate the state vector on the Bloch sphere.

### Unitary

U times U-dagger = I. Every gate is reversible — no information is ever destroyed.

### Acting on a vector

A gate multiplies the state vector: |psi'> = U|psi>.

### Single vs multi-qubit

1-qubit gates are 2x2; 2-qubit gates are 4x4; an n-qubit gate is 2^n x 2^n.

### Universality

A small gate set (e.g. H, T and CNOT) can approximate ANY quantum computation.


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THE THREE PAULI GATES

# The Pauli Gates — X, Y and Z

## Pauli X · NOT

```
X = [[0, 1],
     [1, 0]]

X|0> = |1>
X|1> = |0>
```

A 180-degree rotation about the X axis. The quantum NOT gate — it flips the bit.

## Pauli Y

```
Y = [[0, -i],
     [i,  0]]

Y|0> = i|1>
```

A 180-degree rotation about the Y axis. Combines a bit flip and a phase flip (Y = iXZ).

## Pauli Z · phase flip

```
Z = [[1,  0],
     [0, -1]]

Z|+> = |->
```

A 180-degree rotation about the Z axis. Leaves |0> alone and negates |1> — invisible to measurement alone.


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WHAT TO REMEMBER

# Key Properties of the Pauli Gates

1

### Self-inverse

X.X = Y.Y = Z.Z = I — apply any Pauli twice and you are back where you started.

2

### Hermitian and unitary

Each Pauli is its own conjugate transpose AND its own inverse.

3

### Anticommuting

XY = -YX, YZ = -ZY, ZX = -XZ. Order matters, and swapping it costs a minus sign.

4

### Eigenvalues

Every Pauli has eigenvalues +1 and -1 — the two possible measurement outcomes.

5

### Z eigenstates

|0> and |1> — the computational basis.

6

### X eigenstates

|+> and |-> — the Hadamard basis, used by BB84.


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THE WORKHORSE GATE

# The Hadamard Gate — Making Superposition

## The matrix

```
H = (1/sqrt(2)) x
[[1,  1],
          [1, -1]]
```

The single most important gate. It maps the computational basis to the superposition basis.

## Its action

```
H|0> = |+>
H|1> = |->

H|+> = |0>
```

Creates an equal superposition from a basis state, and converts it back. H is its own inverse: H.H = I.

## On n qubits

```
H on every qubit gives
(1/sqrt(2^n)) sum over all x
of |x>
```

One layer of Hadamards puts the register into a uniform superposition over ALL 2^n inputs — the first step of Deutsch-Jozsa, BV and Grover.


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PARAMETERISED GATES

# Rotation and Phase Gates

## RX, RY, RZ

```
RY(theta) =
[[cos(t/2), -sin(t/2)],
        [sin(t/2),  cos(t/2)]]
```

Continuous rotations about the X, Y and Z axes by a free parameter theta. These are the TRAINABLE gates in quantum machine learning.

## Phase gate P

```
P(phi) = [[1, 0],
                [0, e^(i.phi)]]
```

Applies a relative phase to the |1> component only. Leaves measurement probabilities unchanged in the Z basis.

## S and T

```
S = P(pi/2)
T = P(pi/4)

          S = T.T
```

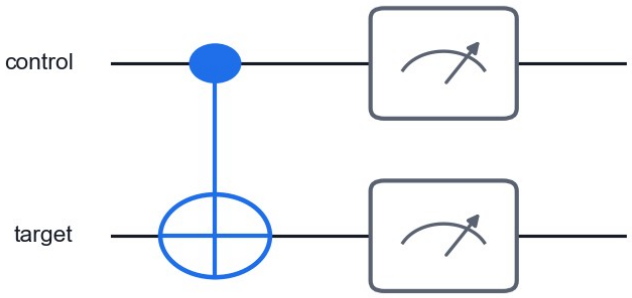
The two standard fixed phase gates. T is the gate that lifts Clifford circuits to universal quantum computation — and the most expensive to make fault-tolerant.


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THE ENTANGLING GATE

# Two-Qubit Gates — CNOT

**The CNOT Gate**




Flip the target iff the control is |1⟩

| INPUT | OUTPUT |
|---|---|
| \|00⟩ | \|00⟩ |
| \|01⟩ | \|01⟩ |
| \|10⟩ | \|11⟩ |
| \|11⟩ | \|10⟩ |

*Flipped rows in teal*

### The rule

Flip the target if and only if the control is |1>.

### Truth table

|00> to |00>, |01> to |01>, |10> to |11>, |11> to |10>.

### It entangles

CNOT applied to a superposed control produces a Bell state — no single-qubit gate can do that.

### Universality

CNOT plus arbitrary single-qubit rotations is a universal gate set.

H on the control, then CNOT — the two-gate recipe for a Bell pair.


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TOPIC 02 · QUICK REFERENCE

# Multi-Qubit Gate Reference

| Gate | Qubits | Action | Key use |
|---|---|---|---|
| **CNOT (CX)** | 2 | Flips the target when the control is \|1> | Entangling; Bell states; oracles |
| **CZ** | 2 | Applies a -1 phase to \|11> only | Grover oracle; phase kickback |
| **Controlled-P** | 2 | Applies e^(i.phi) to \|11> | Quantum Fourier Transform |
| **SWAP** | 2 | Exchanges the states of two qubits | Routing on real hardware |
| **Toffoli (CCX)** | 3 | Flips the target if BOTH controls are \|1> | Reversible classical logic; arithmetic |
| **CCZ** | 3 | Phase-flips \|111> | 3-qubit Grover oracle |


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WHAT TO REMEMBER

# Key Properties — CZ, SWAP and Toffoli

1

### CZ is symmetric

Control and target are interchangeable — CZ(a,b) = CZ(b,a). Unusual for a controlled gate.

2

### CZ from CNOT

H on the target, then CNOT, then H on the target, gives exactly CZ. And vice versa.

3

### SWAP = 3 CNOTs

SWAP is built from three alternating CNOTs — which is why it is expensive on real hardware.

4

### Toffoli is universal

CCNOT alone is universal for REVERSIBLE CLASSICAL computation — it can build AND, NOT and FANOUT.

5

### Toffoli is expensive

Decomposing one Toffoli needs about six CNOTs plus T gates — a major cost in fault-tolerant design.

6

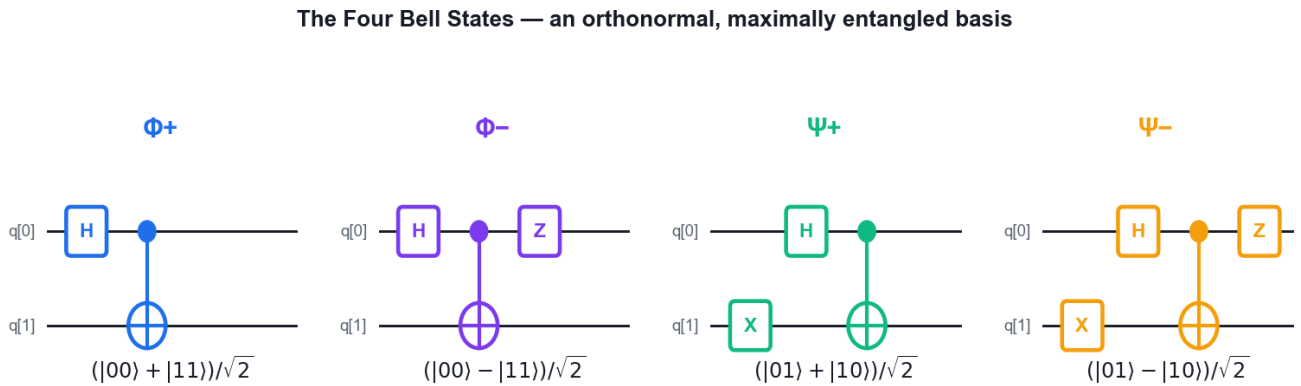
### All are self-inverse

Applying CZ, SWAP or Toffoli twice returns the original state.


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THE FOUR BELL STATES

# Bell States — Maximal Entanglement




**Phi+**

(|00> + |11>)/sqrt(2) — H on q0, then CNOT.

**Phi-**

(|00> - |11>)/sqrt(2) — add Z after the CNOT.

**Psi+**

(|01> + |10>)/sqrt(2) — add X on q1 before the CNOT.

**Psi-**

(|01> - |10>)/sqrt(2) — add both X and Z. The four form an orthonormal basis.

Each Bell state is maximally entangled and mutually orthogonal.


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BEYOND TWO QUBITS

# GHZ States — Entangling n Qubits

1

### The state

(|000> + |111>)/sqrt(2) — all three qubits perfectly correlated.

2

### The recipe

One H on the first qubit, then a CNOT from it to every other qubit.

3

### Generalises

For n qubits: (|0...0> + |1...1>)/sqrt(2), built with one H and n-1 CNOTs.

4

### Fragile

Measuring or losing ANY single qubit destroys the entanglement of the whole state.

5

### Uses

Multi-party QKD, quantum secret sharing, and benchmarking a device's entangling quality.

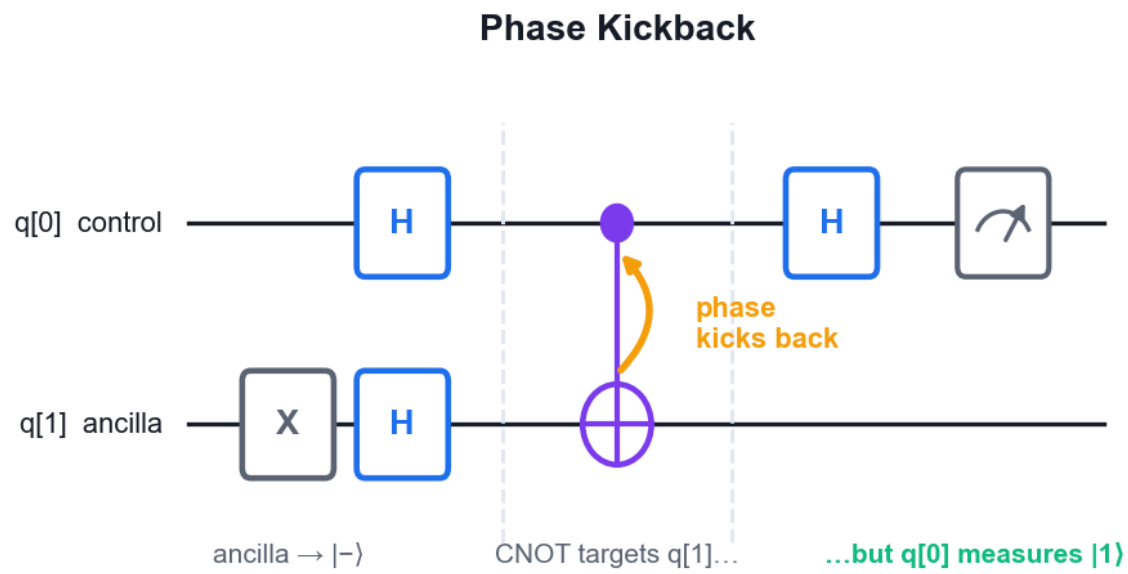
6

### Test of QM

GHZ gives a non-statistical contradiction with local realism — stronger than Bell's.


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# Phase Kickback




$$\text{CNOT}\,|x\rangle|-\rangle = (-1)^x|x\rangle|-\rangle$$ — the answer becomes a PHASE on the control

### The mechanism

A phase applied to the TARGET of a controlled gate appears on the CONTROL instead.

### Why

If the target is an eigenstate of the operation, the controlled gate can only contribute a global-to-relative phase on the control.

### The |-> trick

Put the ancilla in |-> and a CNOT applies a -1 phase exactly when the control is |1>.

### Where it is used

Deutsch, Deutsch-Jozsa, Bernstein-Vazirani, Grover's oracle and Quantum Phase Estimation.

Phase kickback converts an oracle's answer into a measurable phase.


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WHAT YOU'LL DO ON IBM QUANTUM COMPOSER

# Hands-On Labs — Quantum States and Gates

## Lab 4–5

- Single-Qubit Gates — Pauli X, Y, Z and Hadamard
- Rotation and Phase Gates — RX, RY, RZ, S and T

## Lab 6–7

- Two-Qubit Gates — CNOT, CZ, SWAP and Toffoli
- Bell States, GHZ States and Phase Kickback

## —

- —


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TOPIC 02 · HANDS-ON

# Single-Qubit Gates — Pauli X, Y, Z and Hadamard

**LAB 4**

IBM Quantum Composer

Work through the single-qubit gate set on IBM Quantum Composer. Apply each Pauli gate and the Hadamard to |0> and to |1>, read the resulting statevector, and verify from the simulator that each Pauli gate is its own inverse.

1. Open a 1-qubit circuit in IBM Quantum Composer and keep…
2. Apply X to q[0] and confirm the state becomes |1> — X is…
3. Clear, apply Z to q[0], and confirm the state stays |0>
4. Build H then Z
5. Apply Y to |0> and read the statevector: you get i|1>

**YOU'LL PRODUCE** A gate reference table you fill in yourself: the output state and measurement probabilities for X, Y, Z and H on |0> and |1>.

**Full step-by-step guide: Learner Guide, Lab 4**


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TOPIC 02 · HANDS-ON

# Rotation and Phase Gates — RX, RY, RZ, S and T

**LAB 5**

IBM Quantum Composer

Move beyond fixed gates to parameterised rotations. Sweep the angle of an RY gate on IBM Quantum Composer and watch the measurement probability change continuously — the basis of every variational quantum algorithm.

1. Open a 1-qubit circuit with a Measure and place an RY…
2. Set theta = pi/2 and run 1024 shots
3. Set theta = pi and run again
4. Set theta = pi/4 and predict the split BEFORE running…
5. Replace RY with RX(pi) and confirm it also produces |1> —…

**YOU'LL PRODUCE** A probability-vs-angle table for RY(theta) at theta = 0, pi/4, pi/2, 3pi/4 and pi, plus the S and T phase gates demonstrated.

**Full step-by-step guide: Learner Guide, Lab 5**


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TOPIC 02 · HANDS-ON

# Two-Qubit Gates — CNOT, CZ, SWAP and Toffoli

**LAB 6**

IBM Quantum Composer

Build the truth table of the multi-qubit gates by hand on IBM Quantum Composer. Prepare each computational basis input with X gates, apply the gate, and read off the output — the fastest way to make CNOT, CZ, SWAP and Toffoli concrete.

1. Open a 2-qubit circuit
2. Measure both qubits and run
3. Repeat for input |00> (no X gates)
4. Replace the CNOT with a CZ gate on input |11> (X on both…
5. Make the CZ phase visible: build H on q[1], then CZ, then…

**YOU'LL PRODUCE** Completed truth tables for CNOT, SWAP and Toffoli, plus a demonstration that CZ only applies a phase.

**Full step-by-step guide: Learner Guide, Lab 6**


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TOPIC 02 · HANDS-ON

# Bell States, GHZ States and Phase Kickback

**LAB 7**

IBM Quantum Composer

Build all four Bell states and the 3-qubit GHZ state on IBM Quantum Composer, then demonstrate phase kickback — the mechanism that makes the Deutsch, Bernstein-Vazirani and phase-estimation algorithms work.

1. Build the Phi+ Bell state with H on q[0] then CNOT(q[0]…
2. Add an X on q[1] before the CNOT to build Psi+ = (|01> +…
3. Add a Z on q[0] after the CNOT to build Phi- = (|00> -…
4. Extend to three qubits: H on q[0], CNOT(q[0], q[1])…
5. Run the GHZ circuit for 1024 shots and confirm only |000>…

**YOU'LL PRODUCE** The four Bell states, a GHZ state, and a phase-kickback circuit showing the phase moving onto the control qubit.

**Full step-by-step guide: Learner Guide, Lab 7**


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TOPIC 02 RECAP · END OF DAY 1

# Recap — Quantum States and Gates

- Every gate is a unitary matrix, so every quantum operation is reversible.
- X, Y and Z are the Pauli gates — self-inverse, anticommuting, with eigenvalues +1 and -1.
- H creates superposition; RX/RY/RZ rotate by a free angle; S and T apply fixed phases.
- CNOT entangles; CZ phase-flips |11>; SWAP exchanges; Toffoli is reversible classical logic.
- H then CNOT builds a Bell pair. Phase kickback is what makes the query algorithms work.


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03

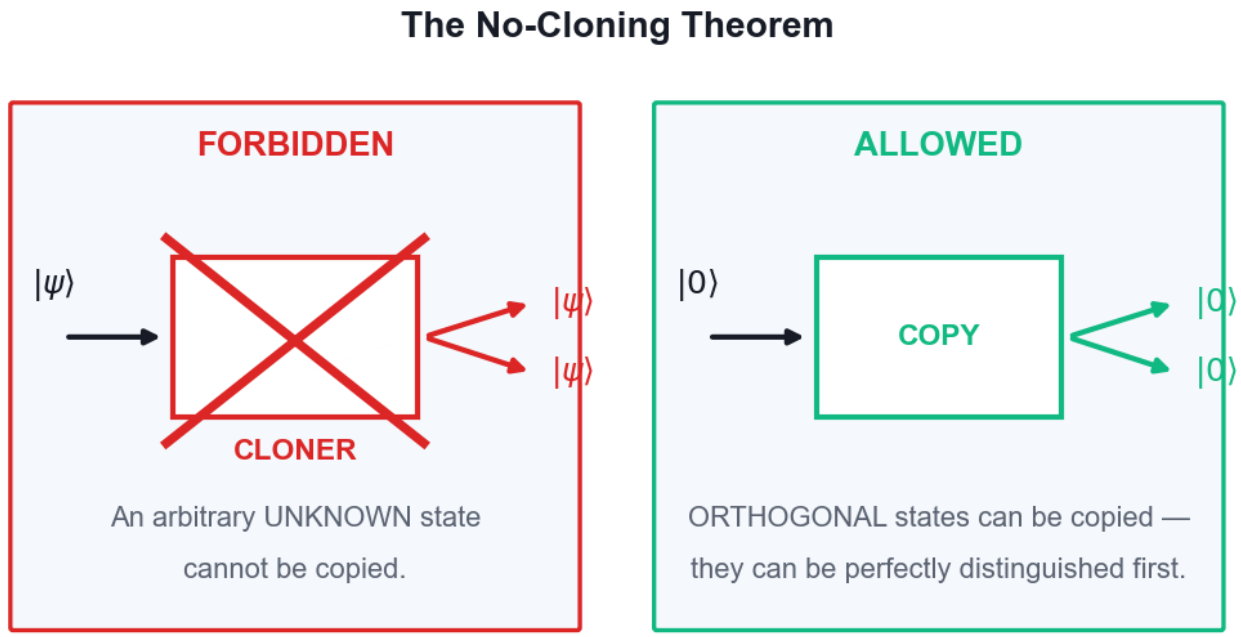
TOPIC 03

# Quantum Algorithms

Coding Quantum Algorithms · Testing Quantum Algorithms


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# The No-Cloning Theorem




### The statement

It is impossible to build a machine that copies an arbitrary UNKNOWN quantum state.

### The proof

Linearity of quantum mechanics forbids it — assume a cloner exists and you derive a contradiction.

### The exception

Orthogonal states CAN be copied, because they can be perfectly distinguished first.

### The consequence

Quantum information cannot be broadcast or backed up — and eavesdroppers cannot copy a key.

No-cloning is why QKD is secure and why teleportation must destroy the original.


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MOVING A QUBIT WITHOUT MOVING A QUBIT

# Quantum Teleportation — The Protocol

1. Alice and Bob share an entangled Bell pair in advance
2. Alice holds the unknown qubit she wants to send
3. Alice performs a Bell measurement on her two qubits
4. Alice sends the two classical bits to Bob
5. Bob applies X and/or Z to recover the exact state


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COMMON MISCONCEPTIONS

# Quantum Teleportation — What It Is and Is Not

1

## The problem

Alice must send an unknown qubit to Bob but can transmit only classical information.

2

## The resources

One shared Bell pair plus exactly two classical bits per qubit teleported.

3

## No faster than light

Bob's qubit is useless until Alice's two classical bits arrive — causality is safe.

4

## The original dies

Alice's measurement destroys her copy — exactly as no-cloning requires.

5

## Not matter transport

Nothing physical travels; only the STATE is transferred onto Bob's existing qubit.

6

## Why it matters

Teleportation underpins quantum repeaters, quantum networks and fault-tolerant gate design.


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TWO CLASSICAL BITS FROM ONE QUBIT

# Superdense Coding — The Mirror Image

1. Alice and Bob share an entangled Bell pair in advance
2. Alice picks her 2-bit message: 00, 01, 10 or 11
3. Alice applies I, X, Z or ZX to HER qubit only
4. Alice sends that single qubit to Bob
5. Bob applies CNOT then H and reads both bits


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TWO SIDES OF ONE COIN

# Teleportation vs Superdense Coding

## Qubit over classical channel

- Quantum teleportation
  - Sends: one QUBIT (unknown state)
  - Consumes: 1 Bell pair + 2 classical bits
  - Alice measures; Bob corrects
  - Use: quantum networks, repeaters
  - Original is destroyed (no-cloning)

## Bits over quantum channel

- Superdense coding
  - Sends: two CLASSICAL BITS
  - Consumes: 1 Bell pair + 1 qubit sent
  - Alice encodes; Bob decodes
  - Use: doubling channel capacity
  - Beats Holevo's bound only via prior entanglement


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# BB84 — Quantum Key Distribution

**BB84 — sift the key by keeping only the matched-basis positions**

| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Alice's bit | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| Alice's basis | Z | X | Z | X | X | Z | X | Z |
| Bob's basis | Z | X | X | X | Z | Z | X | X |
| Bases match? | ✓ | ✓ | × | ✓ | × | ✓ | ✓ | × |
| Sifted key | 0 | 1 | — | 0 | — | 0 | 0 | — |

Shared secret key = 01000 · an eavesdropper injects ~25% errors into these bits

### Bennett & Brassard, 1984

The first and most famous QKD protocol — provably secure by physics, not by hardness.

### Two bases

Alice encodes each bit in the Z basis (|0>, |1>) or the X basis (|+>, |->), chosen at random.

### Sifting

Alice and Bob publicly compare BASES (never bits) and keep only the matched positions.

### Eve is caught

Measuring in the wrong basis disturbs the state, injecting about a 25% error rate in the sifted key.

Security rests on no-cloning and measurement disturbance — not on computational difficulty.


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HOW THE KEY IS AGREED

# The BB84 Protocol Step by Step

1. Alice picks a random bit and a random basis for each qubit
2. Alice sends the encoded qubits over the quantum channel
3. Bob measures each qubit in his own randomly chosen basis
4. They publicly compare bases and discard the mismatches
5. They sacrifice some bits to check the error rate for Eve


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HOW WE MEASURE QUANTUM ADVANTAGE

# The Query (Oracle) Model

1

### The setup

An unknown function f is hidden inside a black box. You may only ASK it about inputs.

2

### The measure

Count the QUERIES, not the gates. This isolates true quantum advantage from implementation detail.

3

### The quantum oracle

A unitary U_f that maps |x>|y> to |x>|y XOR f(x)> — reversible, as every gate must be.

4

### Why XOR

Writing the answer into a second register keeps the oracle reversible and enables phase kickback.

5

### Superposition query

One query on a superposed input evaluates f on ALL inputs at once.

6

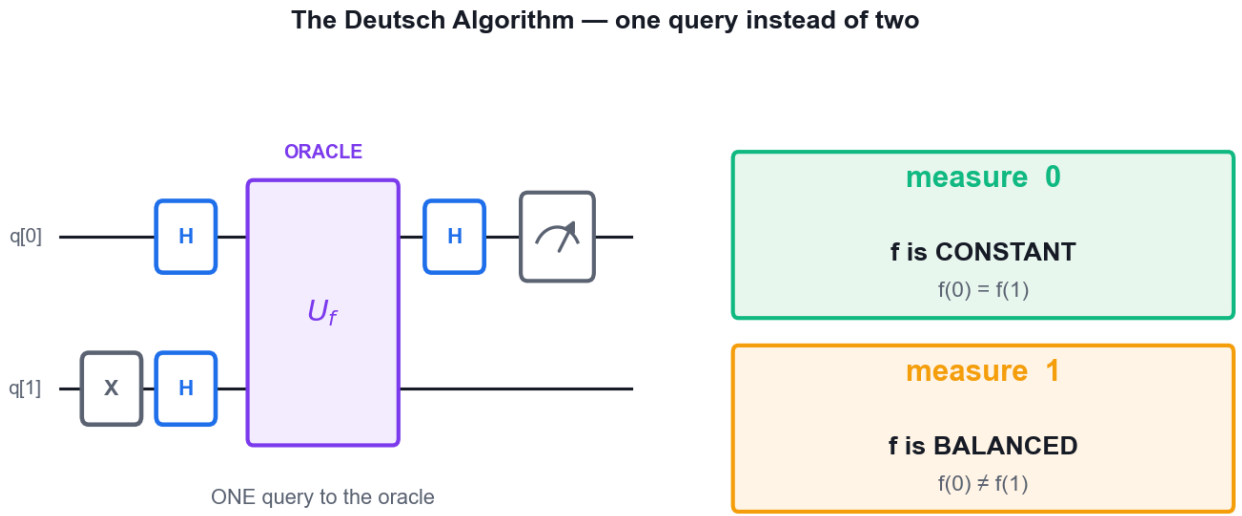
### The catch

Measurement returns only ONE answer — the algorithm must use interference to extract a GLOBAL property.


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THE FIRST QUANTUM SPEEDUP

# The Deutsch Algorithm




### The question

Is f: {0,1} to {0,1} CONSTANT (same output) or BALANCED (different outputs)?

### Classically

You must evaluate f twice — f(0) and f(1) — and compare them.

### Quantum

ONE query. Measure the control: 0 means constant, 1 means balanced.

### The mechanism

Phase kickback writes f(0) XOR f(1) into the control's phase; the final H turns it into a bit.

Two classical queries become one quantum query.


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EXPONENTIAL ADVANTAGE · DETERMINISTIC

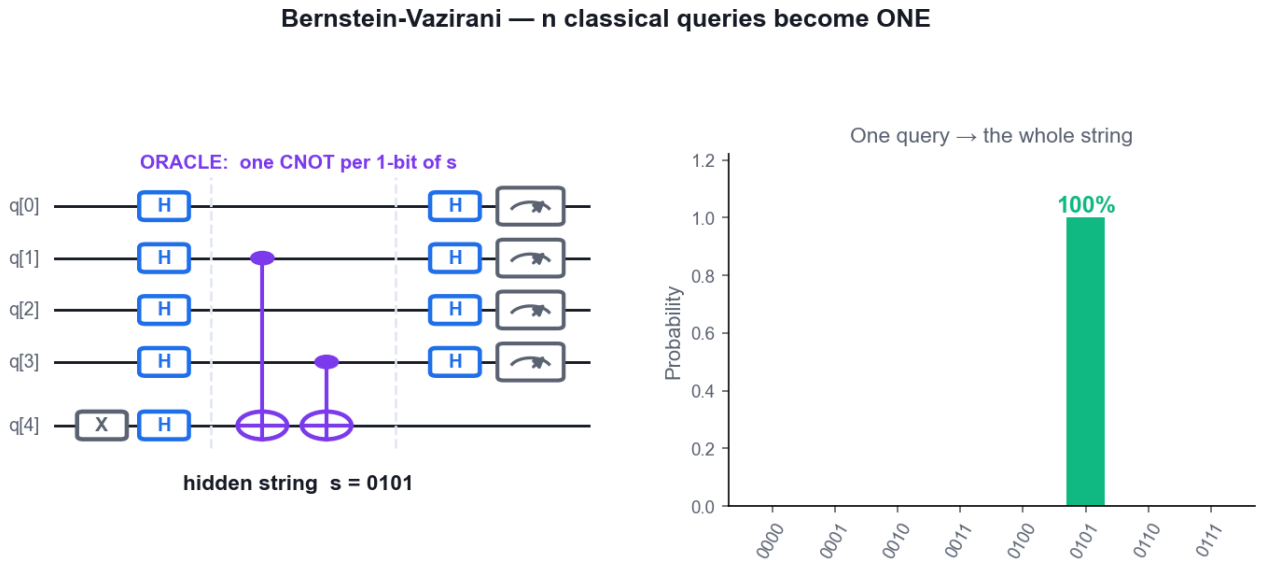
# Deutsch-Jozsa — The Full Speedup

| | Classical (worst case) | Quantum (Deutsch-Jozsa) |
|---|---|---|
| **Queries needed** | 2^(n-1) + 1 | 1 |
| **n = 3** | 5 queries | 1 query |
| **n = 10** | 513 queries | 1 query |
| **n = 20** | 524,289 queries | 1 query |
| **Certainty** | Deterministic | Deterministic |
| **Reading the result** | Compare outputs | All zeros = constant; anything else = balanced |


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n QUERIES BECOME ONE

# The Bernstein-Vazirani Algorithm




The hidden string appears directly in the measurement histogram.

### The problem

f(x) = s.x mod 2 for a hidden n-bit string s. Find s.

### Classically

n queries — probe one bit at a time with 100...0, 010...0 and so on.

### Quantum

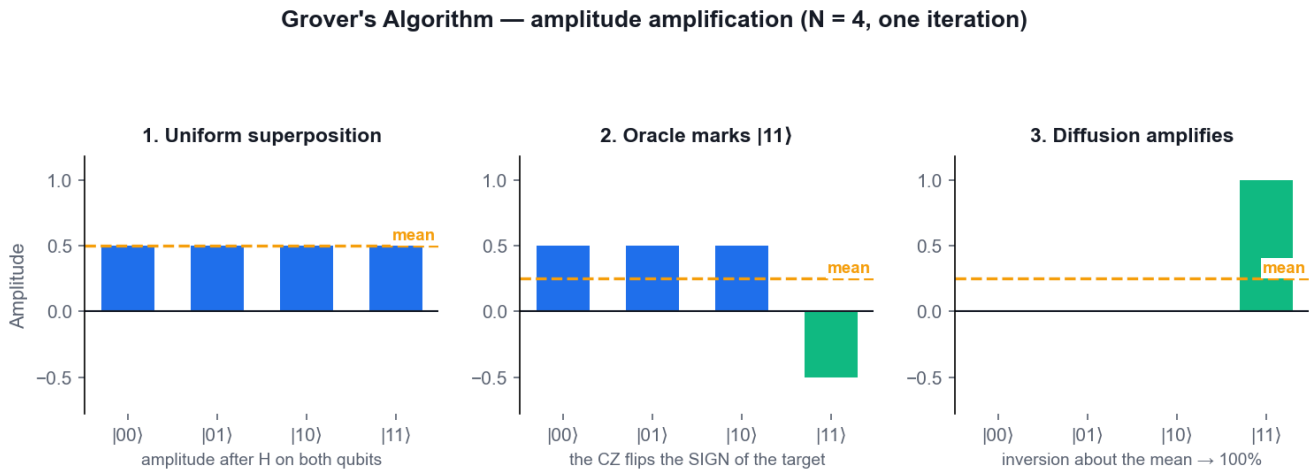
ONE query recovers all n bits of s simultaneously.

### The oracle

One CNOT from each input qubit where s has a 1, onto the |-> ancilla.


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# Grover's Search Algorithm




Amplitude amplification: each iteration rotates the state toward the marked item.

### The problem

Find the one marked item among N unsorted items.

### Classically

N/2 look-ups on average; N in the worst case.

### Quantum

About (pi/4).sqrt(N) iterations — a QUADRATIC speedup.

### Two parts

An ORACLE that phase-flips the target, then a DIFFUSION operator that inverts about the mean.


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AMPLITUDE AMPLIFICATION

# Grover's Algorithm — The Loop

1. Apply H to every qubit for a uniform superposition
2. ORACLE: phase-flip the marked state (e.g. CZ marks |11>)
3. DIFFUSION: H, X, CZ, X, H — invert about the mean
4. Repeat the oracle + diffusion about sqrt(N) times
5. Measure — the marked item appears with high probability


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WHAT TO REMEMBER

# Grover — Practical Points

1

### Iteration count matters

Use about (pi/4).sqrt(N). For N = 4 that is exactly ONE iteration, giving 100%.

2

### Over-rotation hurts

Run too many iterations and the amplitude rotates PAST the target and back down.

3

### Quadratic, not exponential

Grover is a square-root speedup — real, but far weaker than Shor's exponential one.

4

### It is provably optimal

No quantum algorithm can search an unstructured space faster than sqrt(N).

5

### Oracle construction

Mark |01> by putting X gates around the CZ on the qubits that should be 0.

6

### Impact on crypto

Grover halves effective symmetric key strength — AES-128 falls to about 64 bits, so use AES-256.


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THE BASIS OF SHOR'S ALGORITHM

# Quantum Fourier Transform — A Look Ahead

1

### What it is

The quantum analogue of the discrete Fourier transform, acting on amplitudes.

2

### The cost

O(n^2) gates for 2^n amplitudes, versus O(n.2^n) for the classical FFT — exponentially fewer.

3

### The circuit

Hadamards interleaved with controlled-phase rotations, then a final reversal of qubit order.

4

### Where it is used

Quantum Phase Estimation, and hence Shor's factoring algorithm and the HHL linear solver.

5

### The catch

The output amplitudes are not directly readable — you must design interference to extract the answer.

6

### Why it matters

QFT is the source of the exponential speedups, whereas Grover's is only quadratic.


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WHAT YOU'LL DO ON IBM QUANTUM COMPOSER

# Hands-On Labs — Quantum Algorithms

## Lab 8–9

- Quantum Teleportation
- Superdense Coding

## Lab 10–11

- BB84 Quantum Key Distribution
- The Deutsch and Deutsch-Jozsa Algorithms

## Lab 12–13

- The Bernstein-Vazirani Algorithm
- Grover's Search Algorithm


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TOPIC 03 · HANDS-ON

# Quantum Teleportation

**LAB 8**

IBM Quantum Composer

Teleport an unknown qubit state from Alice to Bob on IBM Quantum Composer using one shared Bell pair and two classical bits. Build it until you can do it from memory.

1. Open a 3-qubit circuit in IBM Quantum Composer
2. Prepare the state to be teleported
3. Create the shared Bell pair between Alice and Bob: H on…
4. Add a barrier, then perform Alice's Bell measurement…
5. Measure q[0] and q[1] into the classical bits

**YOU'LL PRODUCE** A 3-qubit teleportation circuit that moves a |+> state from q[0] to q[2] and verifies the transfer.

**Full step-by-step guide: Learner Guide, Lab 8**


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TOPIC 03 · HANDS-ON

# Superdense Coding

**LAB 9**

IBM Quantum Composer

Run teleportation in reverse: send two classical bits by transmitting one qubit. Encode each of the four messages 00, 01, 10 and 11 on IBM Quantum Composer and confirm Bob decodes each one with certainty.

1. Open a 2-qubit circuit
2. Add a barrier
3. To send 00, Alice applies nothing (the identity)
4. Build the decoder that Bob applies: CNOT(q[0], q[1])…
5. Run the 00 circuit for 1024 shots

**YOU'LL PRODUCE** Four superdense-coding circuits, one per two-bit message, each decoding deterministically.

**Full step-by-step guide: Learner Guide, Lab 9**


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TOPIC 03 · HANDS-ON

# BB84 Quantum Key Distribution

**LAB 10**

**IBM Quantum Composer**

Simulate the BB84 key-exchange protocol on IBM Quantum Composer. Encode bits in randomly chosen bases, measure in randomly chosen bases, sift the key — then introduce an eavesdropper and watch the error rate expose her.

1. Understand the encoding: Alice sends bit 0 as |0> or |+>…
2. On a 1-qubit circuit, prepare bit 1 in the X basis: X…
3. Bob measures in the X basis by applying H before the…
4. Now let Bob measure in the WRONG basis: remove his H so…
5. Repeat for at least 8 qubits with a random basis choice…

**YOU'LL PRODUCE** A sifted BB84 key produced without eavesdropping, and the elevated error rate produced when Eve intercepts.

**Full step-by-step guide: Learner Guide, Lab 10**


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TOPIC 03 · HANDS-ON

# The Deutsch and Deutsch-Jozsa Algorithms

**LAB 11**

IBM Quantum Composer

Build the first algorithm that beats every classical method. Determine whether a black-box function is constant or balanced using a single query on IBM Quantum Composer, then scale it up to the multi-qubit Deutsch-Jozsa version.

1. Open a 2-qubit circuit
2. Add a barrier, then build the BALANCED oracle f(x) = x as…
3. Add a barrier, apply H to q[0], and measure q[0] only
4. Run 1024 shots
5. Replace the oracle with the CONSTANT function f(x) = 0 —…

**YOU'LL PRODUCE** Four Deutsch circuits (both constant and both balanced oracles) and a 3-qubit Deutsch-Jozsa circuit.

**Full step-by-step guide: Learner Guide, Lab 11**


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TOPIC 03 · HANDS-ON

# The Bernstein-Vazirani Algorithm

**LAB 12**

IBM Quantum Composer

Recover a hidden 4-bit string in a single query instead of four. Build the circuit for the hidden string 0101 (decimal 5) until it is second nature.

1. Open a 5-qubit circuit: q[0]-q[3] are the input register…
2. Prepare the ancilla in |->: apply X to q[4], then H to…
3. Apply H to every input qubit q[0], q[1], q[2] and q[3] to…
4. Add a barrier, then build the oracle for the hidden…
5. Add a barrier and apply H again to all four input qubits…

**YOU'LL PRODUCE** A 5-qubit Bernstein-Vazirani circuit that reads out the hidden string 0101 deterministically.

**Full step-by-step guide: Learner Guide, Lab 12**


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TOPIC 03 · HANDS-ON

# Grover's Search Algorithm

**LAB 13**

**IBM Quantum Composer**

Build the best-known quantum search algorithm on IBM Quantum Composer. Mark a target item with an oracle, amplify it with the diffusion operator, and find it in O(sqrt(N)) steps.

1. Open a 2-qubit circuit with 2 classical bits
2. Add a barrier, then build the ORACLE that marks |11> with…
3. Add a barrier, then build the DIFFUSION operator…
4. Measure both qubits and run 1024 shots
5. Change the marked item to |01>: put an X on q[0] before…

**YOU'LL PRODUCE** A 2-qubit Grover circuit that finds the marked item with near-certainty, plus a 3-qubit version.

**Full step-by-step guide: Learner Guide, Lab 13**


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TOPIC 03 RECAP

# Recap — Quantum Algorithms

- No-cloning forbids copying an unknown state — the basis of QKD security.
- Teleportation moves a qubit with one Bell pair and two classical bits; superdense coding is the mirror image.
- BB84 exchanges a provably secure key; an eavesdropper shows up as roughly 25% errors in the sifted key.
- Deutsch, Deutsch-Jozsa and Bernstein-Vazirani all beat classical query counts using phase kickback.
- Grover finds a marked item in about sqrt(N) steps — a quadratic, provably optimal speedup.


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04

TOPIC 04

# Quantum Computing Applications

Introduction to Quantum Error Correction · Introduction to Quantum Machine Learning


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# Why Quantum Error Correction?

1

### Qubits are fragile

They decohere in microseconds — heat, vibration and stray fields all destroy the state.

2

### Gates are imperfect

A two-qubit gate error of 1% compounds fast: 100 gates and the answer is noise.

3

### No copying allowed

No-cloning forbids the classical trick of simply keeping three backup copies.

4

### No peeking allowed

Measuring the data qubit to check it would COLLAPSE the superposition you are protecting.

5

### The solution

Encode one LOGICAL qubit across many PHYSICAL qubits, and measure only the PARITY.

6

### The prize

Below the error threshold, more qubits make the logical error rate arbitrarily small.


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THE SAME IDEA, HARDER CONSTRAINTS

# Classical vs Quantum Error Correction

## Bits

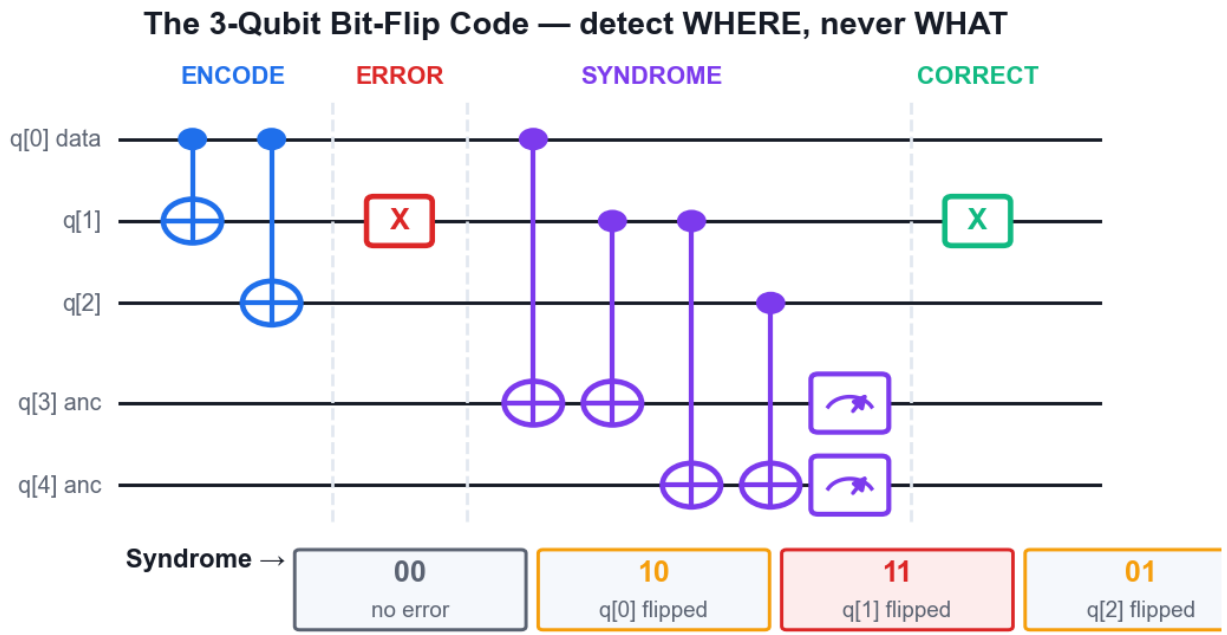
- Classical repetition code
  - Copy the bit three times: 0 becomes 000
  - A flip gives 010 — take the majority vote
  - You may freely read and copy bits
  - Only ONE error type exists: the bit flip

## Qubits

- Quantum error correction
  - Entangle into 3 qubits — you cannot copy
  - Measure PARITY via ancillas, never the data
  - Reading the data would collapse it
  - TWO error types: bit flip (X) and phase flip (Z)


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# The 3-Qubit Bit-Flip Code




Syndrome 00 = no error, 10 = q0, 11 = q1, 01 = q2.

## Encode

Two CNOTs spread |psi> across three qubits: logical |0> = |000>, logical |1> = |111>.

## Error

A stray X flips one qubit — say the middle one, giving |010>.

## Syndrome

Two ancillas measure the parities Z0Z1 and Z1Z2 — revealing WHERE, never WHAT.

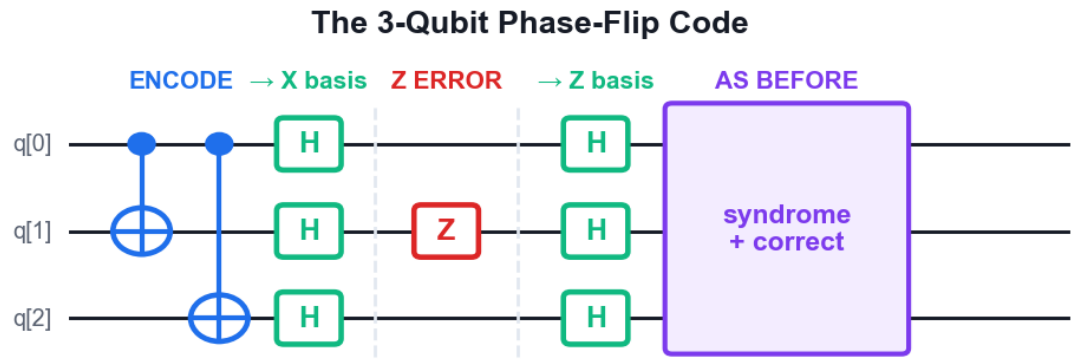
## Correct

Syndrome 11 means qubit 1 flipped; apply X to qubit 1 and the logical state is restored.


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CORRECTING A Z ERROR

# The 3-Qubit Phase-Flip Code




The trick: $HZH = X$ — conjugating by Hadamards turns a PHASE flip into a BIT flip

### The blind spot

A Z error is completely INVISIBLE to the bit-flip code — the syndrome reads 00.

### The trick

H Z H = X. Conjugating by Hadamards turns a phase flip into a bit flip.

### The code

Encode as before, then apply H to all three qubits — logical states become |+++> and |--->.

### Then reuse

Sandwich the bit-flip machinery between Hadamard layers and the same syndrome works.

Every quantum error is a combination of X and Z — correct both and you correct all.


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# Quantum Error-Correcting Codes

| Code | Physical qubits | Corrects | Notes |
|---|---|---|---|
| **Bit-flip code** | 3 | One X error | Teaching code; blind to phase errors |
| **Phase-flip code** | 3 | One Z error | The bit-flip code conjugated by Hadamards |
| **Shor code** | 9 | ANY single-qubit error | The first full QEC code (1995); 3 blocks of 3 |
| **Steane code** | 7 | ANY single-qubit error | CSS code from the Hamming code; more efficient |
| **Surface code** | ~1,000 per logical | ANY single-qubit error | 2-D lattice, high threshold — the leading candidate |


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# Fault-Tolerant Quantum Computing

1

### The definition

Errors do not spread uncontrollably, and correction is built into EVERY step of the computation.

2

### Logical qubits

One useful qubit encoded in many physical ones — roughly 1,000 to 1 for the surface code.

3

### Transversal gates

Apply a logical gate as separate physical gates so a single error cannot cascade.

4

### Syndrome measurement

Stabiliser checks detect errors continuously without ever touching the logical state.

5

### Threshold theorem

Below a physical error rate of about 0.1–1%, arbitrarily long computation becomes possible.

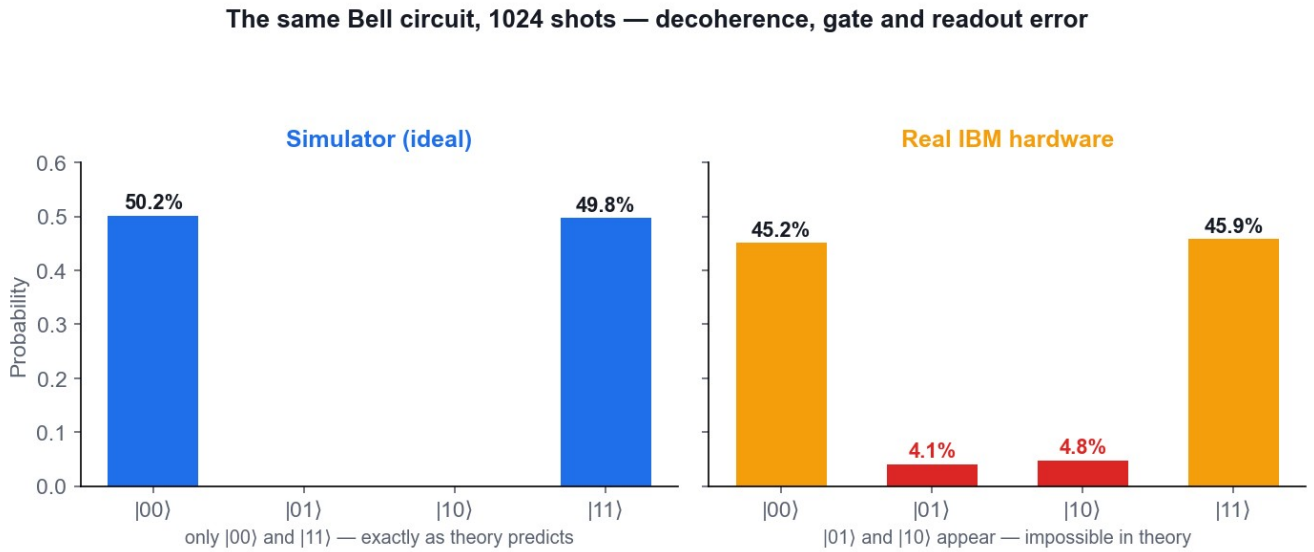
6

### The gap

Useful fault tolerance needs millions of physical qubits — today's machines have hundreds.


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# Simulator vs Real Hardware




This difference is exactly what Lab 16 asks you to explain.

### On the simulator

Perfect qubits, no decoherence, no noise — the Bell state gives ONLY |00> and |11>.

### On real hardware

|01> and |10> appear too — outcomes that are theoretically impossible.

### The causes

T1 relaxation, T2 dephasing, two-qubit gate infidelity, readout error and crosstalk.

### The mitigations

Shallower circuits, better-calibrated qubits, more shots, readout mitigation — then QEC.


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# Introduction to Quantum Machine Learning

1

### The hybrid loop

A quantum circuit computes; a CLASSICAL optimiser updates its parameters. Repeat until converged.

2

### Data encoding

Classical data is loaded as rotation angles (angle encoding) or amplitudes (amplitude encoding).

3

### The ansatz

A parameterised circuit of trainable RY/RZ rotations interleaved with entangling CNOTs.

4

### Measurement

The expectation value of an observable becomes the model's output, feeding a cost function.

5

### VQE

Variational Quantum Eigensolver — finds a molecule's ground-state energy. Chemistry's killer app.

6

### QAOA

Quantum Approximate Optimisation Algorithm — tackles combinatorial problems such as MaxCut.


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HOW A QML MODEL IS TRAINED

# The Hybrid Quantum-Classical Training Loop

1. Encode the classical input data as rotation angles
2. Run the parameterised ansatz on the quantum device
3. Measure to obtain an expectation value
4. A classical optimiser computes the cost and gradient
5. Update the parameters and repeat until converged


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CURRENT STATUS AND OUTLOOK

# Where We Are Today — The NISQ Era

1

### NISQ

Noisy Intermediate-Scale Quantum: 50–1,000+ qubits, high error rates, short coherence.

2

### Not yet fault-tolerant

IBM, Google and IonQ machines cannot yet run Shor's algorithm at useful scale.

3

### Already useful

Optimisation pilots, molecular simulation, certified randomness and QKD deployments.

4

### Next 5–10 years

Better qubits, early logical qubits, quantum accelerators alongside GPUs in the cloud.

5

### The challenges

Scaling to a million qubits, correction overhead, millikelvin cryogenics and control wiring.

6

### Act now

Post-quantum cryptography migration and workforce skills should start today — harvest-now-decrypt-later.


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WHAT YOU'LL DO ON IBM QUANTUM COMPOSER

# Hands-On Labs — Quantum Computing Applications

## Lab 14–15

- Quantum Error Correction — The Bit-Flip Code
- Quantum Error Correction — The Phase-Flip Code

## Lab 16–17

- Simulator vs Real Quantum Hardware
- Introduction to Quantum Machine Learning

## —

- —


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TOPIC 04 · HANDS-ON

# Quantum Error Correction — The Bit-Flip Code

**LAB 14**

IBM Quantum Composer

Protect one logical qubit against a bit-flip error by encoding it into three physical qubits. Deliberately inject an X error on IBM Quantum Composer, detect it with a syndrome measurement, and correct it.

1. Open a 5-qubit circuit: q[0]-q[2] hold the encoded…
2. ENCODE: place the logical state on q[0] (use H for a…
3. Add a barrier, then INJECT an error: place a single X…
4. Add a barrier, then measure the first syndrome: CNOT from…
5. Measure the second syndrome: CNOT from q[1] to q[4] and…

**YOU'LL PRODUCE** A 3-qubit bit-flip code that detects and corrects an injected X error on any one of the three qubits.

**Full step-by-step guide: Learner Guide, Lab 14**


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TOPIC 04 · HANDS-ON

# Quantum Error Correction — The Phase-Flip Code

**LAB 15**

IBM Quantum Composer

Phase flips are invisible to the bit-flip code. Learn the standard trick — surround the code with Hadamards to turn Z errors into X errors — and correct a phase flip on IBM Quantum Composer.

1. First show the problem: build the bit-flip code from Lab…
2. Now build the phase-flip code
3. Add a barrier and INJECT a Z error on q[1] — a phase flip…
4. Add a barrier, then apply H to all three code qubits again
5. Run the same syndrome extraction as Lab 14: CNOTs from…

**YOU'LL PRODUCE** A 3-qubit phase-flip code that detects and corrects an injected Z error.

**Full step-by-step guide: Learner Guide, Lab 15**


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TOPIC 04 · HANDS-ON

# Simulator vs Real Quantum Hardware

**LAB 16**

IBM Quantum Composer, IBM…

Run the same circuit on the ideal simulator and on real IBM quantum hardware, compare the histograms, and diagnose the difference.

1. Rebuild the Bell-state circuit from Lab 3: H on q[0]…
2. Run it on the SIMULATOR with 1024 shots and record the…
3. Switch the backend to a real IBM quantum device from the…
4. Wait for the job to complete in the queue, then open the…
5. Compare: the hardware histogram still peaks at |00> and…

**YOU'LL PRODUCE** A side-by-side comparison of the Bell-state histogram on the simulator and on real hardware, with the error sources identified.

**Full step-by-step guide: Learner Guide, Lab 16**


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# Introduction to Quantum Machine Learning

**LAB 17**

IBM Quantum Composer

Build a parameterised variational circuit — the building block of every quantum machine learning model. Vary the parameters on IBM Quantum Composer, observe the effect on the output distribution, and see how a hybrid quantum-classical training loop works.

1. Understand the pattern: a variational circuit has a…
2. Open a 2-qubit circuit
3. Add the ENTANGLING layer: a CNOT with control q[0] and…
4. Add the TRAINABLE layer: an RY gate on each qubit with…
5. Measure both qubits and run 1024 shots

**YOU'LL PRODUCE** A 2-qubit variational ansatz whose rotation angles you tune to steer the output distribution towards a target.

**Full step-by-step guide: Learner Guide, Lab 17**


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TOPIC 04 RECAP

# Recap — Quantum Computing Applications

- QEC encodes one logical qubit across many physical ones and measures parity, never the data itself.
- The bit-flip code catches X errors; conjugating with Hadamards catches Z errors; Shor and Steane catch both.
- Fault tolerance needs the physical error rate below threshold — and roughly 1,000 physical qubits per logical one.
- Real hardware produces outcomes that are impossible in theory, caused by decoherence, gate and readout error.
- QML runs a parameterised circuit inside a classical optimisation loop — VQE for chemistry, QAOA for optimisation.


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WRAP-UP

# Course Summary & Next Steps


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LEARNING OUTCOMES

# What You Achieved

1

## LO1 · Platforms

Used IBM Quantum Composer and the Bloch sphere to test gates, circuits and algorithms.

2

## LO2 · Gates & logic

Applied Pauli, Hadamard, rotation, phase, CNOT, CZ, SWAP and Toffoli gates to derive solutions.

3

## LO3 · Algorithms

Built and tested teleportation, superdense coding, BB84, Deutsch-Jozsa, BV and Grover.

4

## LO4 · Applications

Diagnosed hardware noise, corrected quantum errors and trained a variational QML circuit.


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# Summary — The Whole Course on One Slide

1

## Superposition

A qubit holds |0> and |1> at once; H is the gate that creates it.

2

## Entanglement

H then CNOT builds a Bell pair — correlations no classical system can reproduce.

3

## Interference

Algorithms make wrong answers cancel and right answers reinforce.

4

## Phase kickback

Turns an oracle's hidden answer into a measurable phase on the control register.

5

## Speedups

Deutsch-Jozsa and BV are exponential in queries; Grover is quadratic and provably optimal.

6

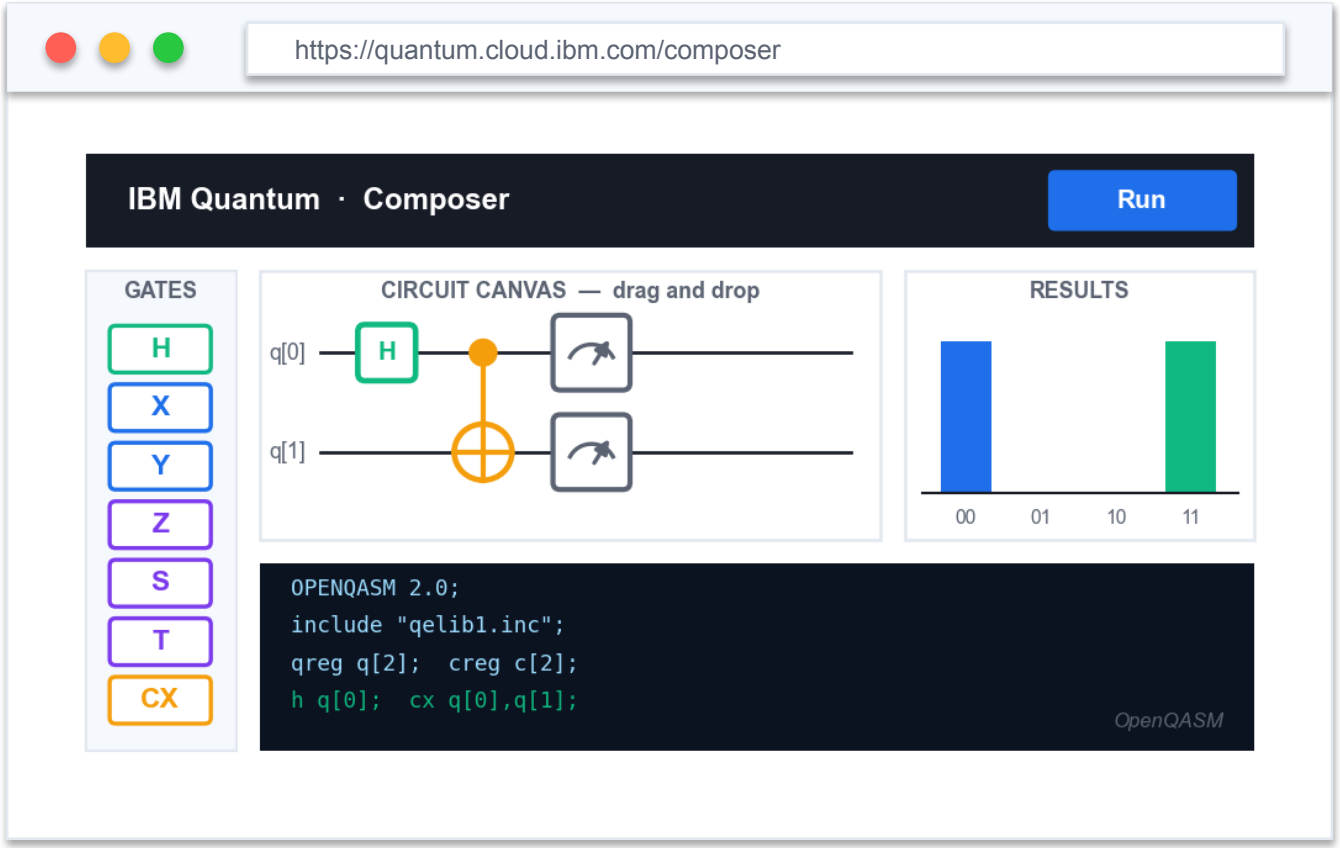
## Reality check

Real devices are noisy — QEC and fault tolerance are the road to useful computation.


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KEEP BUILDING AFTER THE COURSE

# Keep Practising on IBM Quantum




1. Rebuild Labs 8, 12 and 13 from a blank canvas — these are the core algorithm circuits.
2. Re-run each lab and predict the histogram BEFORE you press Run.
3. Re-target Grover's oracle to each of |00>, |01>, |10> and |11>.
4. Change the Bernstein-Vazirani hidden string and confirm the readout follows.
5. Run one circuit on real hardware and explain the difference.

Your IBM Quantum account stays free after the course — keep building.


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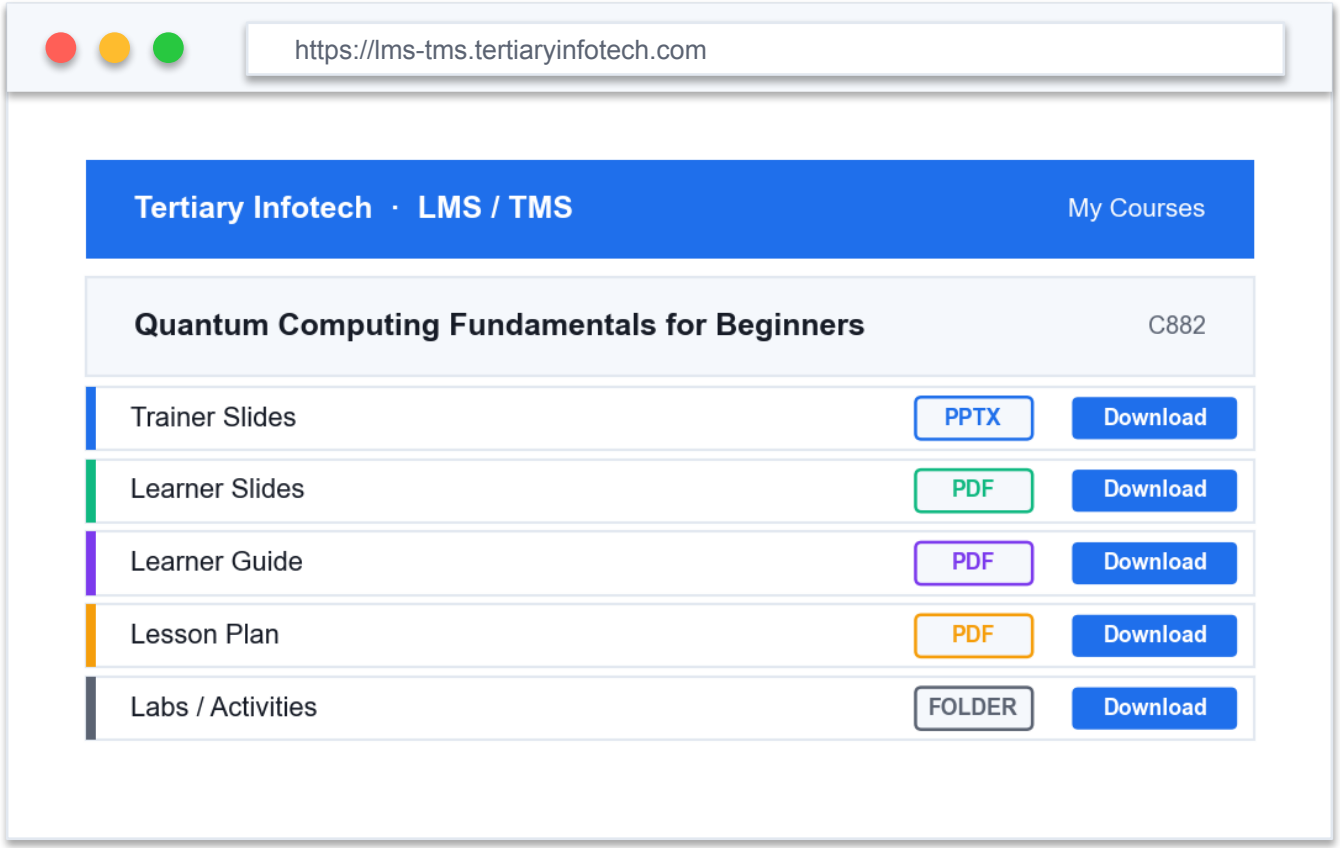
# Continue Your Learning

1. Securing the Future with Quantum Computing and Cryptography
2. Python Fundamental Course for Beginners
3. Building Your First Machine Learning Model with Python and Tensorflow
4. Practical Reinforcement Learning for Beginners
5. Practical Image Generation Using GAN, VAE, and Diffusion Models


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BEFORE YOU LEAVE

# Course Feedback




1. Sign in to the LMS/TMS portal.
2. Complete the course-feedback form for this course.
3. Share what worked best and what needs more time.
4. Your e-certificate of completion is issued after the course.
5. Download it from the portal once released.

Your feedback shapes the next run of this course.


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WE'RE HERE IF YOU NEED US

# Support

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END OF COURSE

# Thank You!

You can now build, run and diagnose quantum circuits on IBM Quantum Composer — and explain why they work.


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